Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

EIN 4360 Facilities Planning and Material Handling

Module 12– Facility Location Models: Minisum

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POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

Types of Models

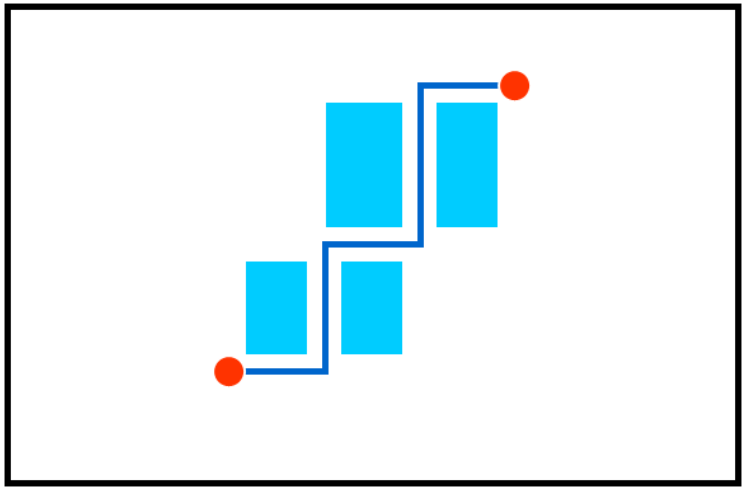
- **Single-Facility Location Model**
- Location-allocation Model

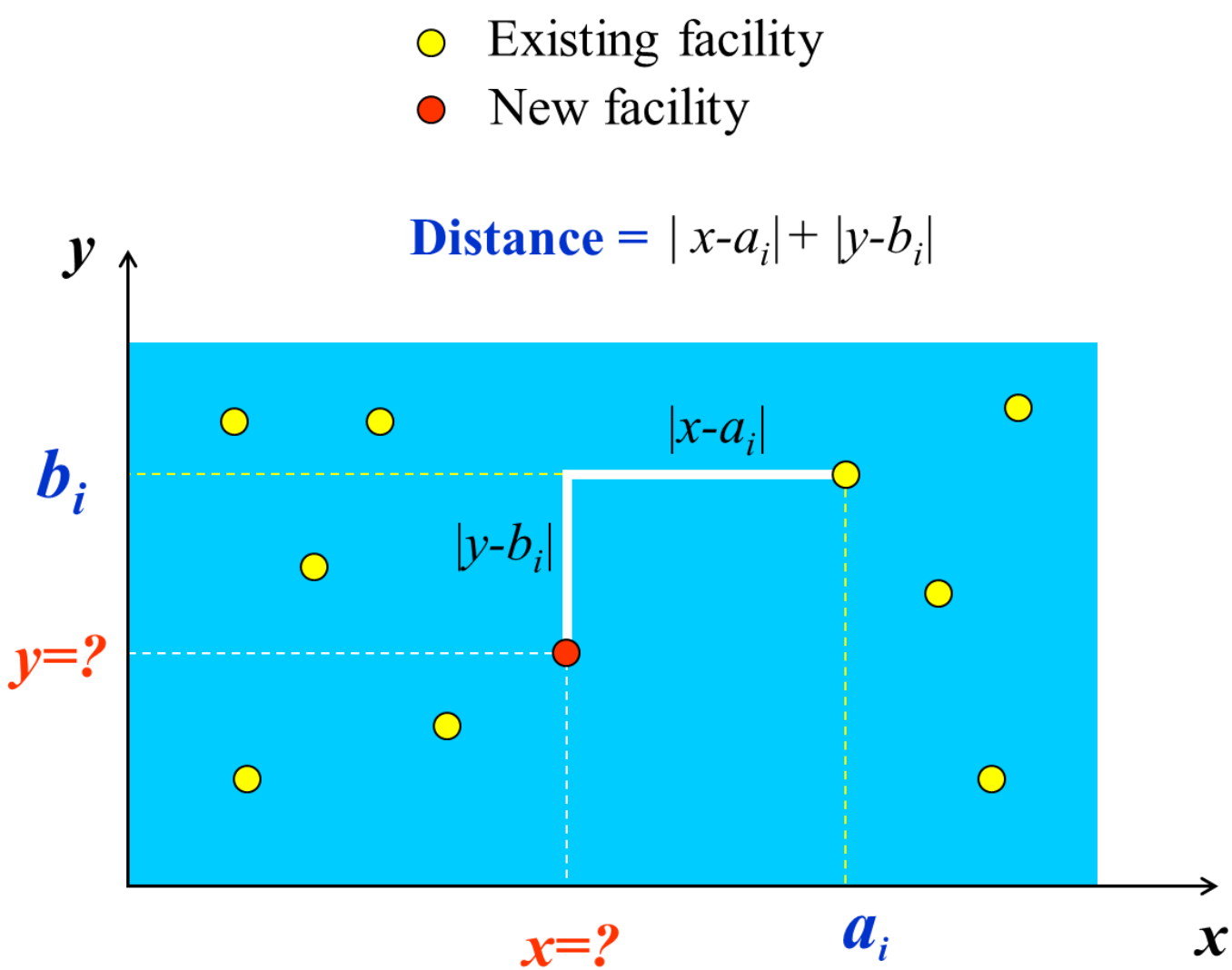
Single-Facility Location Model

- **Minisum Location Problem**
  - Minimizing the sum of the weighted distance between the new facility and other existing facilities
- **Minimax Location Problem**
  - Minimizing the maximum distance between the new facility and any existing facility

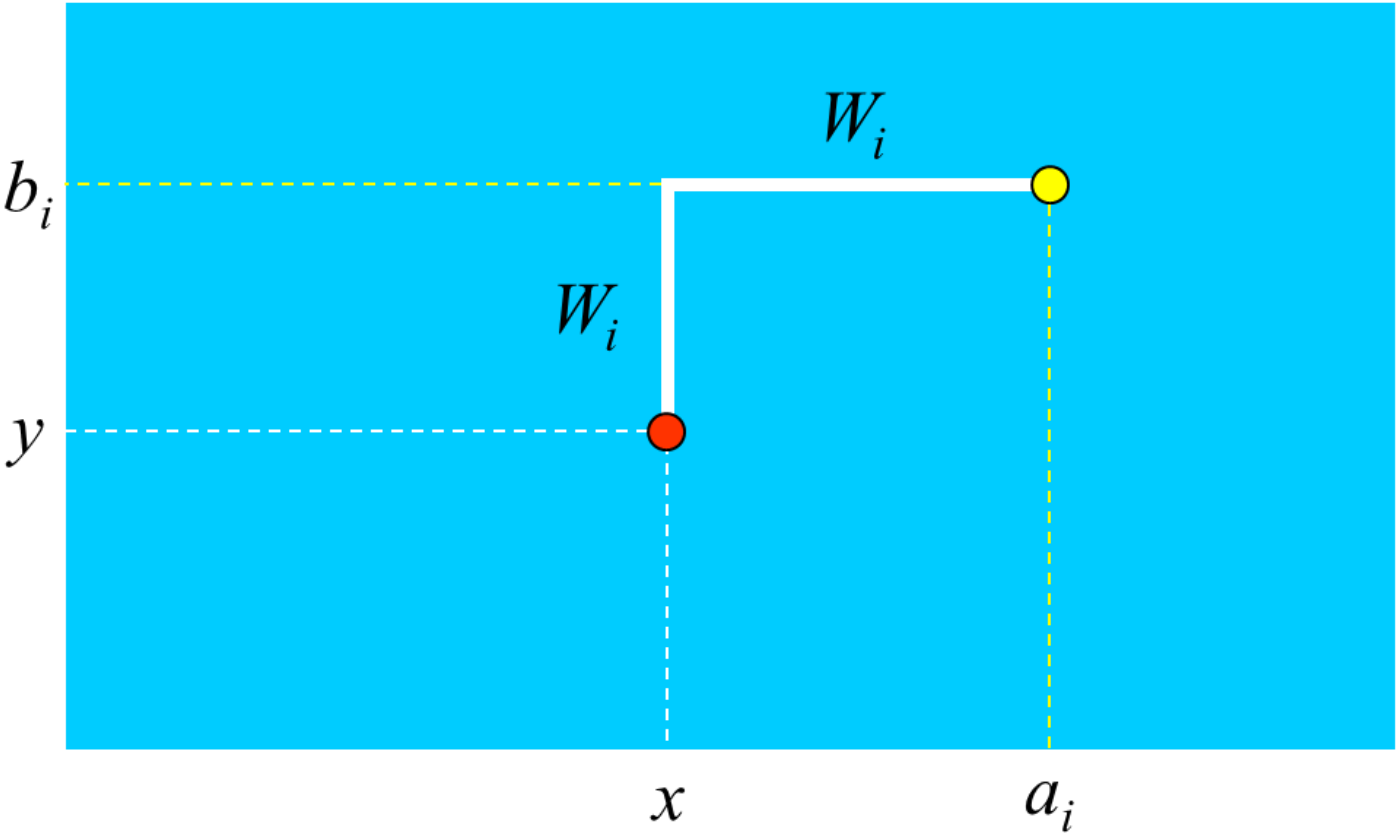
Distances

- **Rectilinear distances**

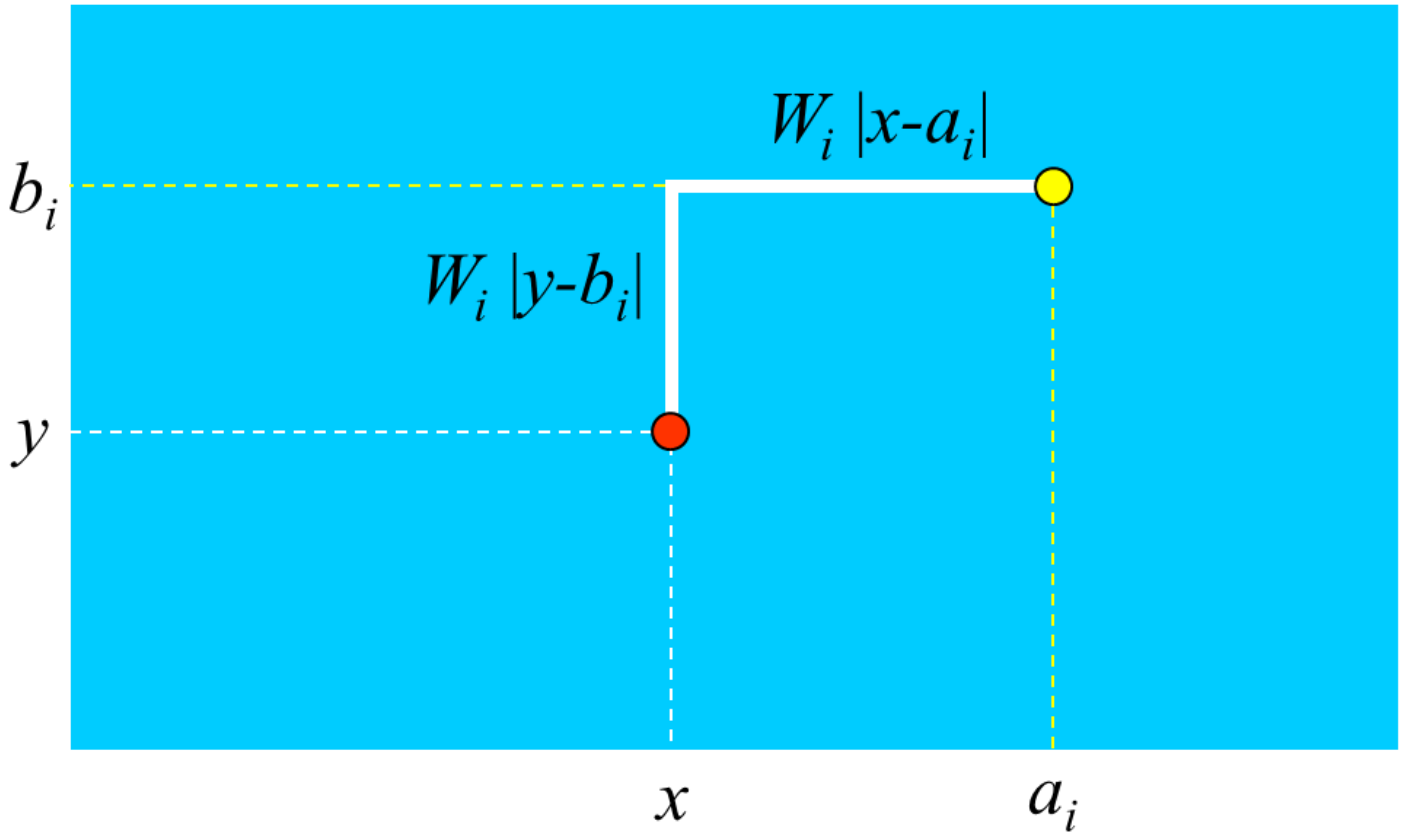



Existing facility
New facility
Distance = $|x-a_i| + |y-b_i|$
y
b_i
$|x-a_i|$
$|y-b_i|$
y=?
x=?
a_i
x

W = Number of trips from ● to ●



$$Weighted\ Distance = W_i\,|x\text{-}a_i| + W_i\,|y\text{-}b_i|$$



Minisum Location Problem with Rectilinear Distances

$$Minimize\ f(x,y) = \sum_{i=1}^{m} (W_i|x-a_i| + W_i|y-b_i|)$$

$$Minimize\ f(x,y) = \sum_{i=1}^{m} W_i|x-a_i| + \sum_{i=1}^{m} W_i|y-b_i|$$

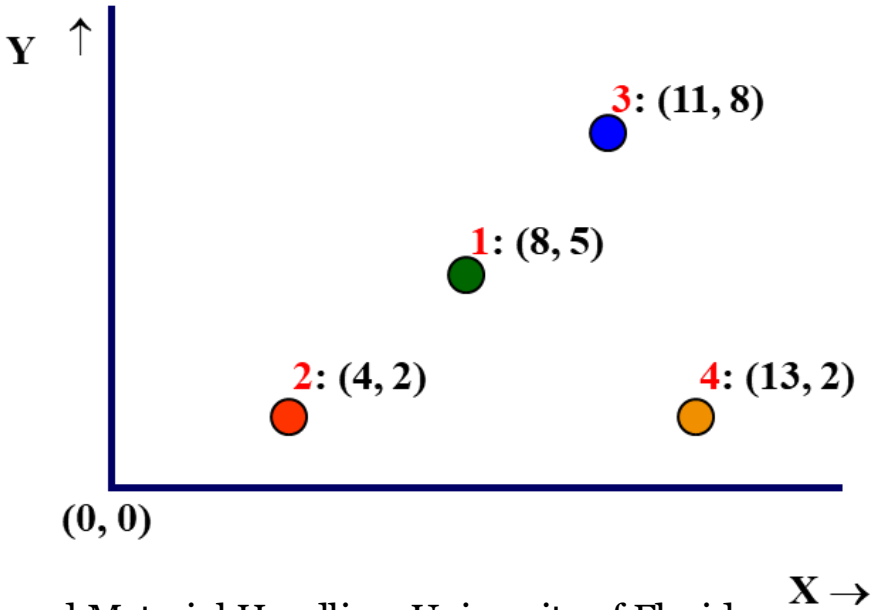
$$Minimize\ f(x) = \sum_{i=1}^{m} W_i|x-a_i|$$

$$Minimize\ f(y) = \sum_{i=1}^{m} W_i|y-b_i|$$

Separable models

Example

Facilities 1, 2, 3, and 4 are located at the points (8, 5), (4, 2), (11, 8) and (13, 2), respectively. It is estimated that there will be 9, 6, 4, and 12 trips per week, respectively, between the facilities and a proposed new facility. Find the location of the new facility.



Example

	x-coordinate	*y-coordinate*	*trips*
Facility 1	$a_1 = 8$	$b_1 = 5$	$W_1 = 9$
Facility 2	$a_2 = 4$	$b_2 = 2$	$W_2 = 6$
Facility 3	$a_3 = 11$	$b_3 = 8$	$W_3 = 4$
Facility 4	$a_4 = 13$	$b_4 = 2$	$W_4 = 12$

$$\textit{Minimize } f(x,y) = 9\,|x-8| + 6\,|x-4| + 4\,|x-11| + 12\,|x-13| + 9\,|y-5| + 6\,|y-2| + 4\,|y-8| + 12\,|y-2|$$

Reformulation as an LP Model

$$\text{Minimize } f(x,y) = 9\,|x-8| + 6\,|x-4| + 4\,|x-11| + 12\,|x-13| + 9\,|y-5| + 6\,|y-2| + 4\,|y-8| + 12\,|y-2|$$

$$\text{Minimize } Z_1 = 9(r_1+s_1) + 6(r_2+s_2) + 4(r_3+s_3) + 12(r_4+s_4)$$

$$x - r_1 + s_1 = 8$$
$$x - r_2 + s_2 = 4$$
$$x - r_3 + s_3 = 11$$
$$x - r_4 + s_4 = 13$$
$$r_i, s_i \geq 0,\ i = 1,2,3,4$$

$$|x - a_i| = r_i + s_i$$
$$x - a_i = r_i - s_i$$
$$r_i, s_i \geq 0$$

$$\text{Minimize } Z_2 = 9(r_1+s_1) + 6(r_2+s_2) + 4(r_3+s_3) + 12(r_4+s_4)$$

$$y - r_1 + s_1 = 5$$
$$y - r_2 + s_2 = 2$$
$$y - r_3 + s_3 = 8$$
$$y - r_4 + s_4 = 2$$
$$r_i, s_i \geq 0,\ i = 1,2,3,4$$

$$|y - b_i| = r_i + s_i$$
$$y - b_i = r_i - s_i$$
$$r_i, s_i \geq 0$$

Reformulation as an LP Model

Minimize 9r1 + 9s1 + 6r2 + 6s2 + 4r3 + 4s3 + 12r4 + 12s4

s.t.

x - r1 + s1 = 8
x - r2 + s2 = 4
x - r3 + s3 = 11
x - r4 + s4 = 13
r1 >=0
s1 >=0
r2 >=0
s2 >=0
r3 >=0
s3 >=0
r4 >=0
s4 >=0

END

Reformulation as an LP Model

Minimize 9r1 + 9s1 + 6r2 + 6s2 + 4r3 + 4s3 + 12r4 + 12s4

s.t.

y - r1 + s1 = 5
y - r2 + s2 = 2
y - r3 + s3 = 8
y - r4 + s4 = 2
r1 >=0
s1 >=0
r2 >=0
s2 >=0
r3 >=0
s3 >=0
r4 >=0
s4 >=0

END

Reformulation as an LP Model

LP OPTIMUM FOUND AT STEP 6

OBJECTIVE FUNCTION VALUE

1) 93.00

VARIABLE	VALUE	REDUCED COST
r1	3.000000	0.000000
s1	0.000000	18.000000
r2	7.000000	0.000000
s2	0.000000	12.000000
r3	0.000000	7.000000
s3	0.000000	1.000000
r4	0.000000	24.000000
s4	2.000000	0.000000
x	11.0000	0.000000

ROW	SLACK OR SURPLUS	DUAL PRICES
2)	0.000000	9.000000
3)	0.000000	6.000000
4)	0.000000	-3.000000
5)	0.000000	-12.000000
6)	3.000000	0.000000
7)	0.000000	0.000000
8)	7.000000	0.000000
9)	0.000000	0.000000
10)	0.000000	0.000000
11)	0.000000	0.000000
12)	0.000000	0.000000
13)	2.000000	0.000000

Reformulation as an LP Model

Optimal solution:

$(x^*, y^*) = (11, 2)$

Cost:

$Z_1 + Z_2 = 93 + 51$

$= 144$

LP OPTIMUM FOUND AT STEP 4

OBJECTIVE FUNCTION VALUE

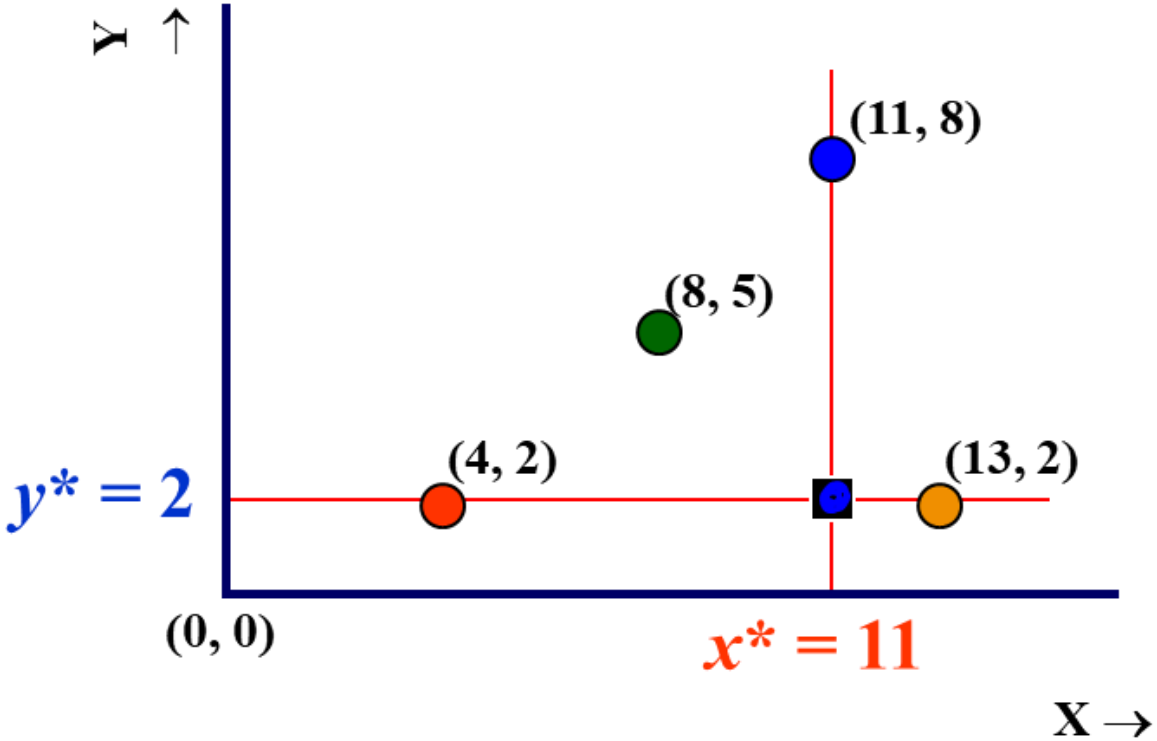
1) **51.00000**

VARIABLE	VALUE	REDUCED COST
R1	0.000000	18.000000
S1	3.000000	0.000000
R2	0.000000	5.000000
S2	0.000000	7.000000
R3	0.000000	8.000000
S3	6.000000	0.000000
R4	0.000000	0.000000
S4	0.000000	24.000000
Y	**2.000000**	0.000000

ROW	SLACK OR SURPLUS	DUAL PRICES
2)	0.000000	-9.000000
3)	0.000000	1.000000
4)	0.000000	-4.000000
5)	0.000000	12.000000
6)	0.000000	0.000000
7)	0.000000	0.000000
8)	0.000000	0.000000
9)	0.000000	0.000000
10)	3.000000	0.000000
11)	0.000000	0.000000
12)	6.000000	0.000000
13)	0.000000	0.000000

NO. ITERATIONS= 4

Optimal Location



Minisum: 4 Solution Approach

1. Solvers (just seen)
2. Cumulative Weight Algorithm
3. Cumulative Weight Diagram
4. Resultant Force Diagram

} **Graphical Approaches**

Cumulative Weight Algorithm

To Find x^*

To Find y^*

Facility 1	$a_1 = 8$	$b_1 = 5$	$W_1 = 9$
Facility 2	$a_2 = 4$	$b_2 = 2$	$W_2 = 6$
Facility 3	$a_3 = 11$	$b_3 = 8$	$W_3 = 4$
Facility 4	$a_4 = 13$	$b_4 = 2$	$W_4 = 12$

Cumulative Weight Algorithm

Rules to determine the optimal value of *x*

- Arrange the a_i's in increasing order
- Compute *sum* of weights (trips) for each value
- Optimal *x* is the value at which the *cumulative* weight $\geq$ ½ (sum of weights)

Rules to determine the optimal value of *y*

- Arrange the b_i's in increasing order
- Compute *sum* of weights (trips) for each value
- Optimal *y* is the value at which the *cumulative* weight $\geq$ ½ (sum of weights)

Calculation of Optimal x*

Facility i	Coordinate a_i	Trips w_i	Sum of w_i
2	4	6	6
1	8	9	15
3	11	4	19
4	13	12	31

x^*

$$\frac{1}{2}\sum_{i=1}^{4} W_i = \frac{31}{2} = 15.5$$

Calculation of Optimal y*

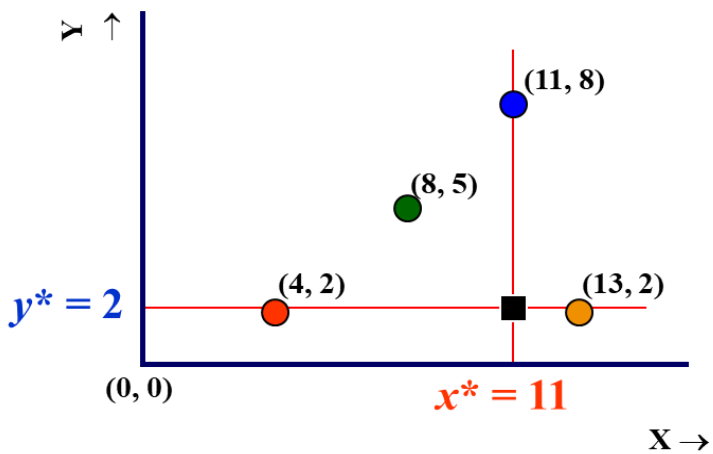
$$\frac{1}{2}\sum_{i=1}^{4} W_i = \frac{31}{2} = 15.5$$

Facility i	*Coordinate* b_i	*Weight* w_i	*Sum of* w_i
2	2	6	6
4	2	12	18
1	5	9	27
3	8	4	31

y^*

Optimal Location: $(x^*, y^*) = (11, 2)$

Cumulative Weight Algorithm



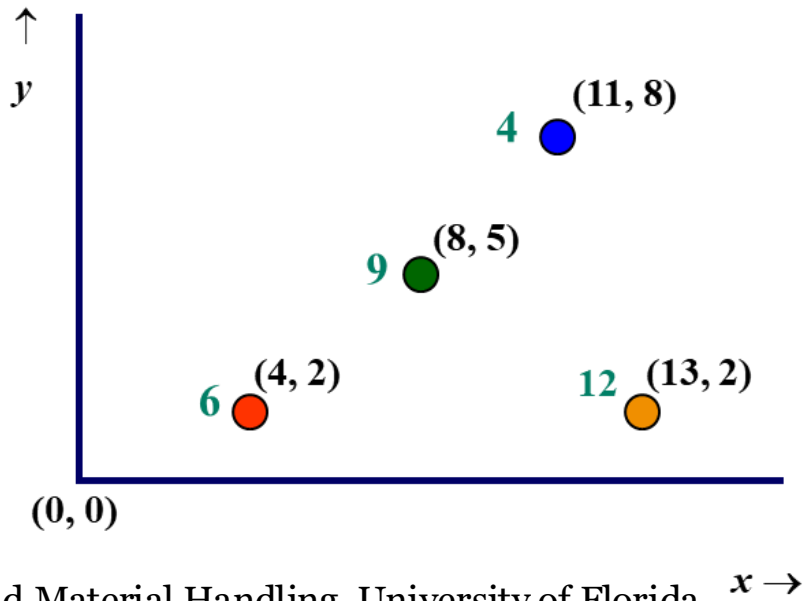
$$f(x,y) = 9\,|\,11-8\,| + 6\,|\,11-4\,| + 4\,|\,11-11\,| + 12\,|\,11-13\,| +$$
$$9\,|\,2-5\,| + 6\,|\,2-2\,| + 4\,|\,2-8\,| + 12\,|\,2-2\,|$$

$$= 93 + 51$$

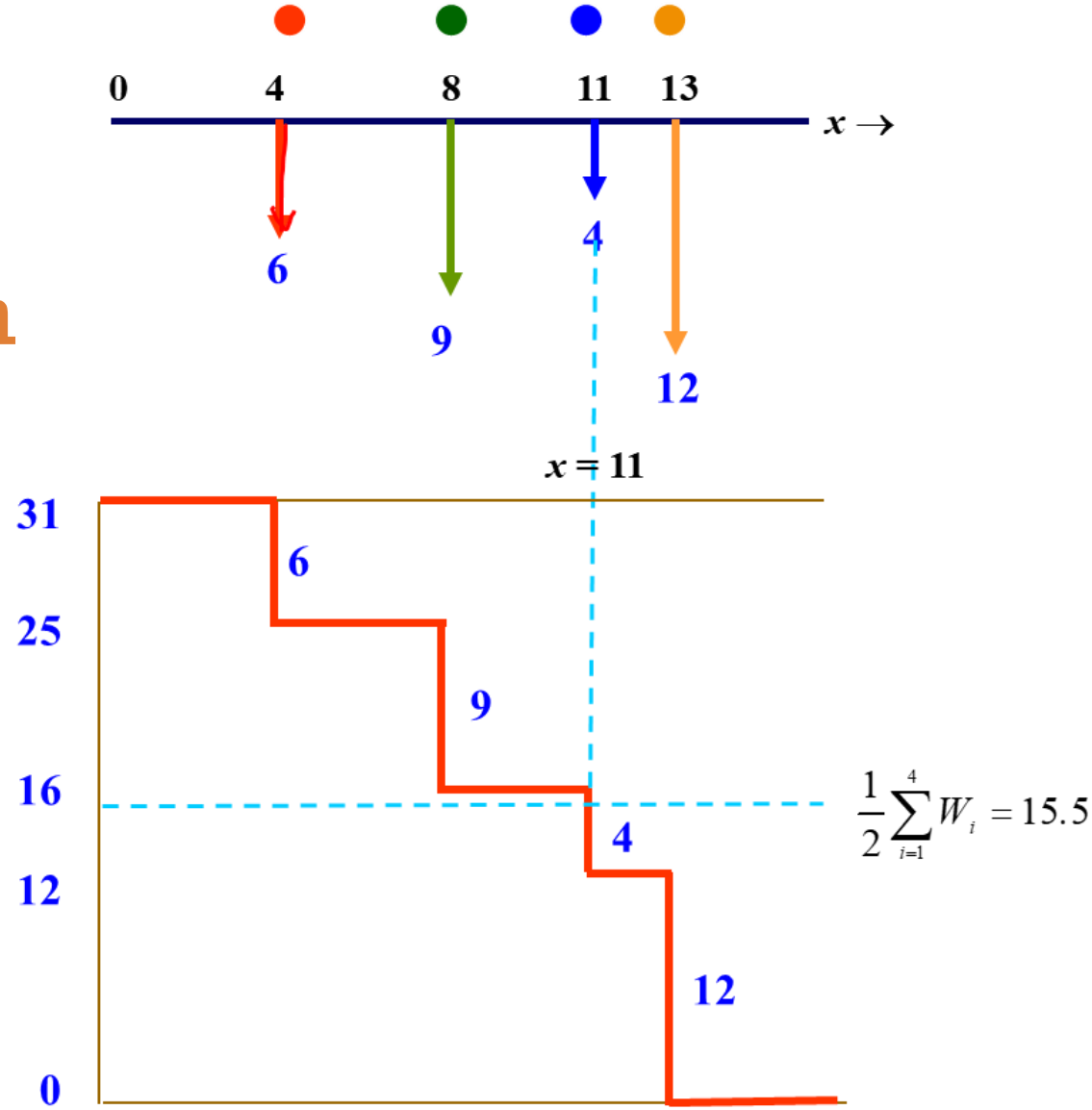
$$= 144$$

Example

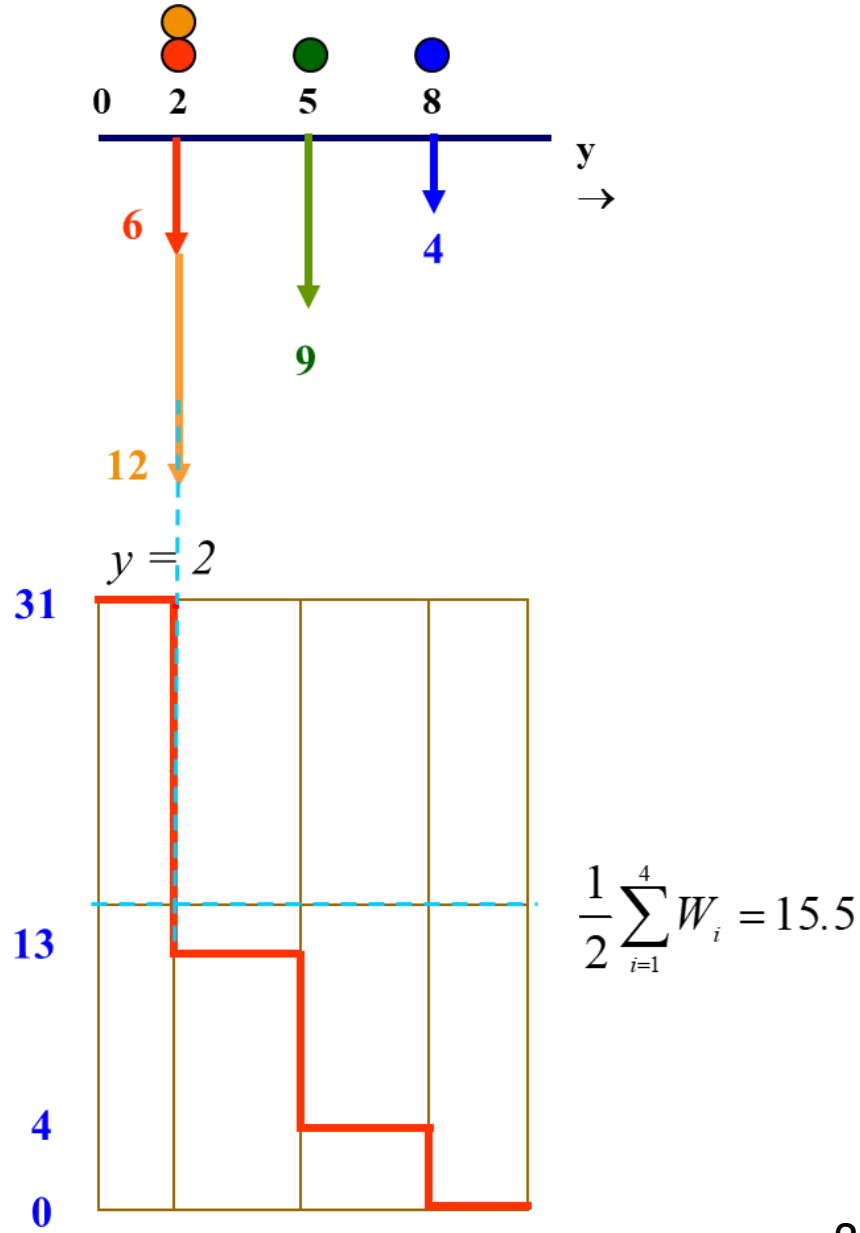
Facilities 1, 2, 3, and 4 are located at the points (8, 5), (4, 2), (11, 8) and (13, 2), respectively. It is estimated that there will be 9, 6, 4, and 12 trips per week, respectively, between the facilities and a proposed new facility. Find the location of the new facility.



Example: Cumulative Weight Diagram for **x**

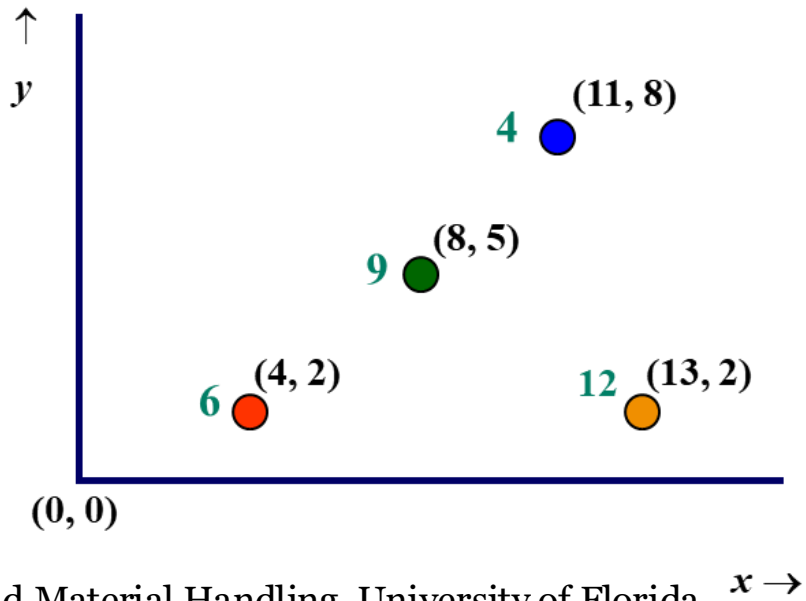


Example: Cumulative Weight Diagram for **y**



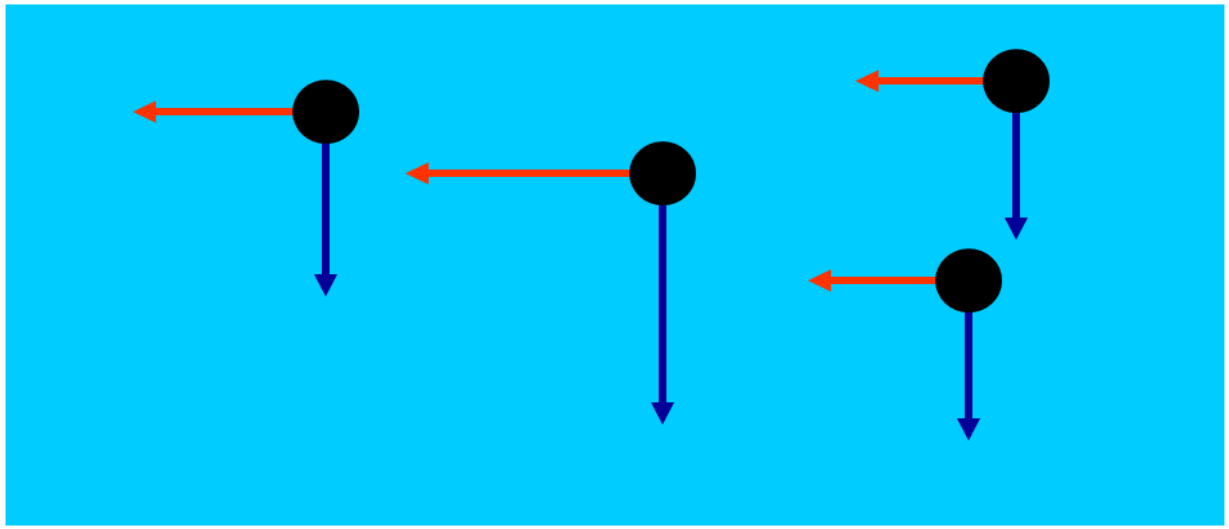
Example

Facilities 1, 2, 3, and 4 are located at the points (8, 5), (4, 2), (11, 8) and (13, 2), respectively. It is estimated that there will be 9, 6, 4, and 12 trips per week, respectively, between the facilities and a proposed new facility. Find the location of the new facility.

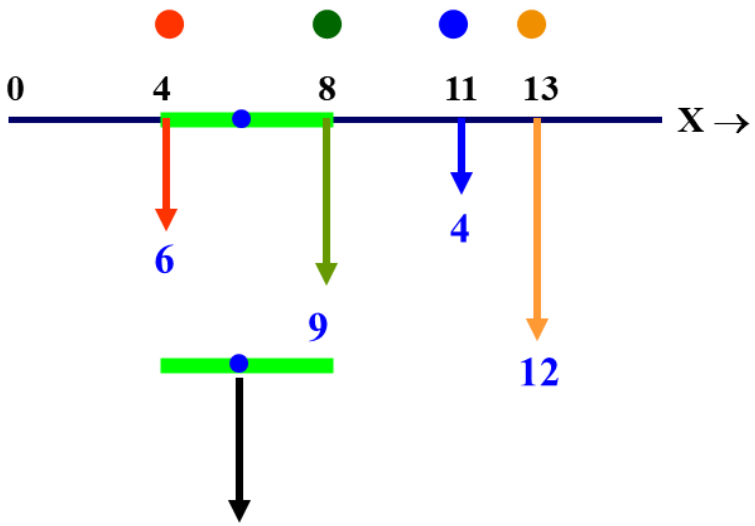


Resultant Force Diagram

Key point: the number of **trips** is viewed as a **force**. If we are finding ***x*** this force acts **horizontally**. If we are finding ***y*** this force acts **vertically**.

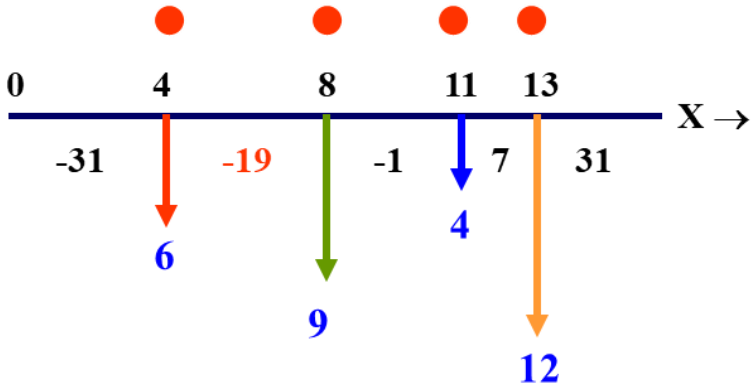


Resultant Force Diagram

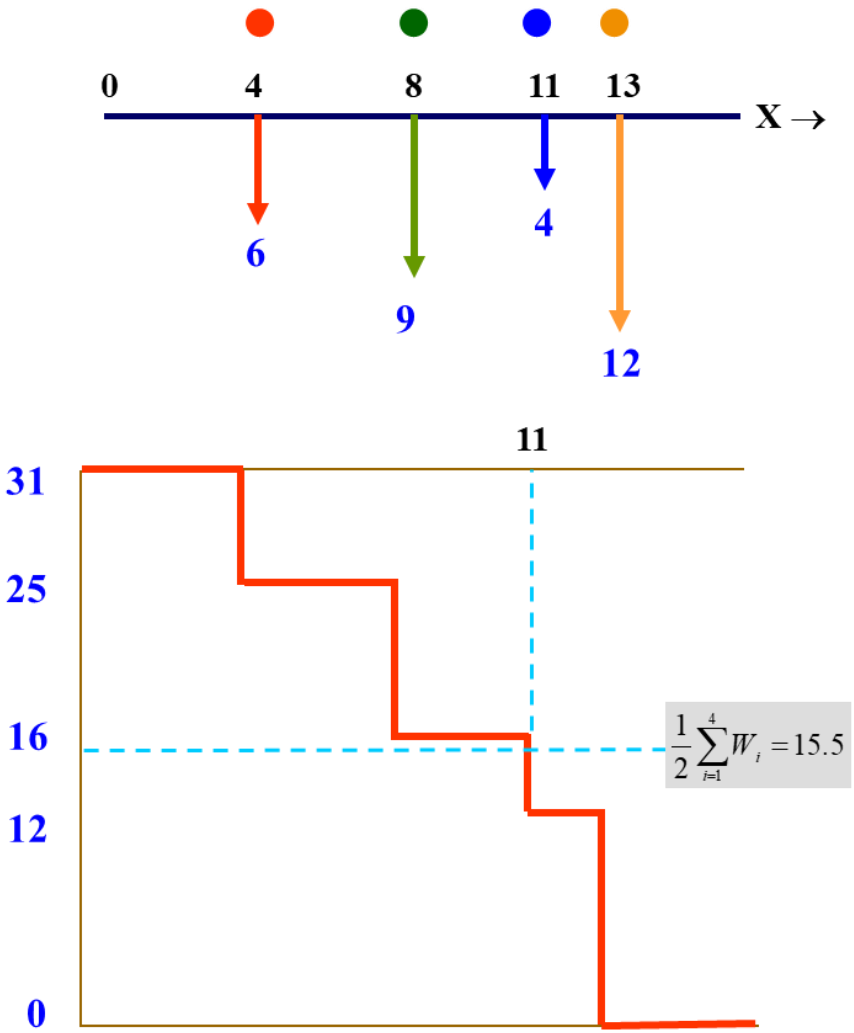


Resultant Force = Total Force $_{left}$ – Total Force $_{right}$

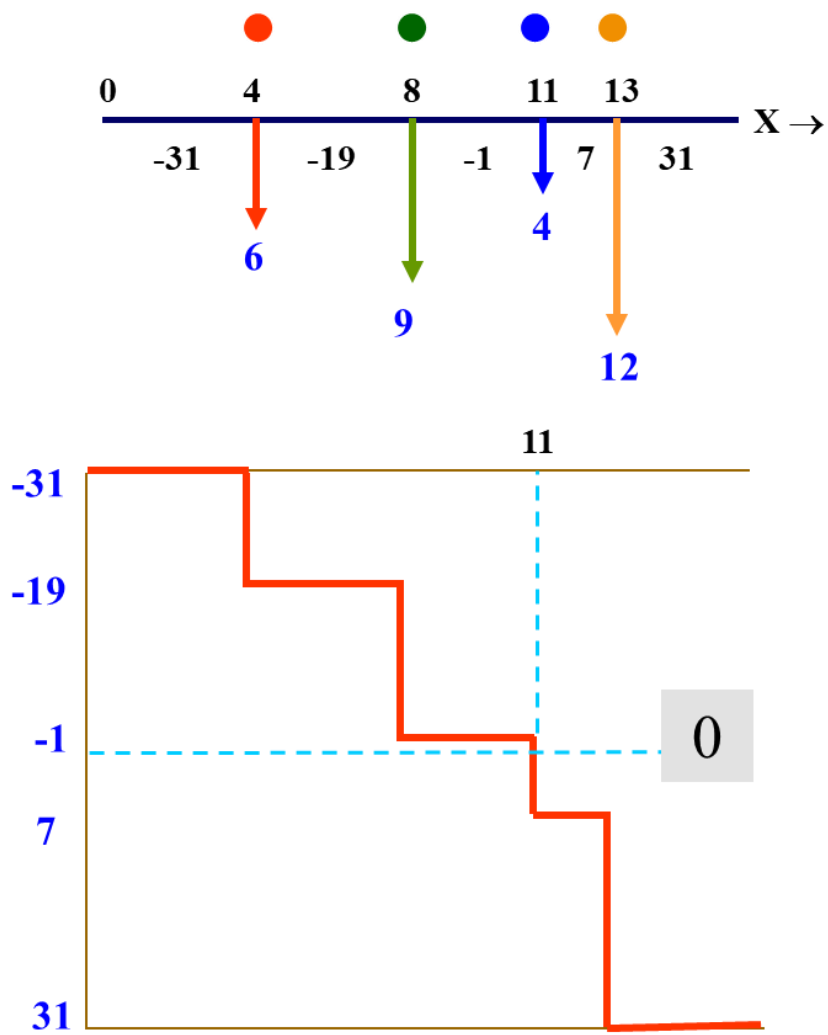
RF = 6 – (9+4+12) = -19



Cumulative Weight Diagram



Resultant Force Diagram



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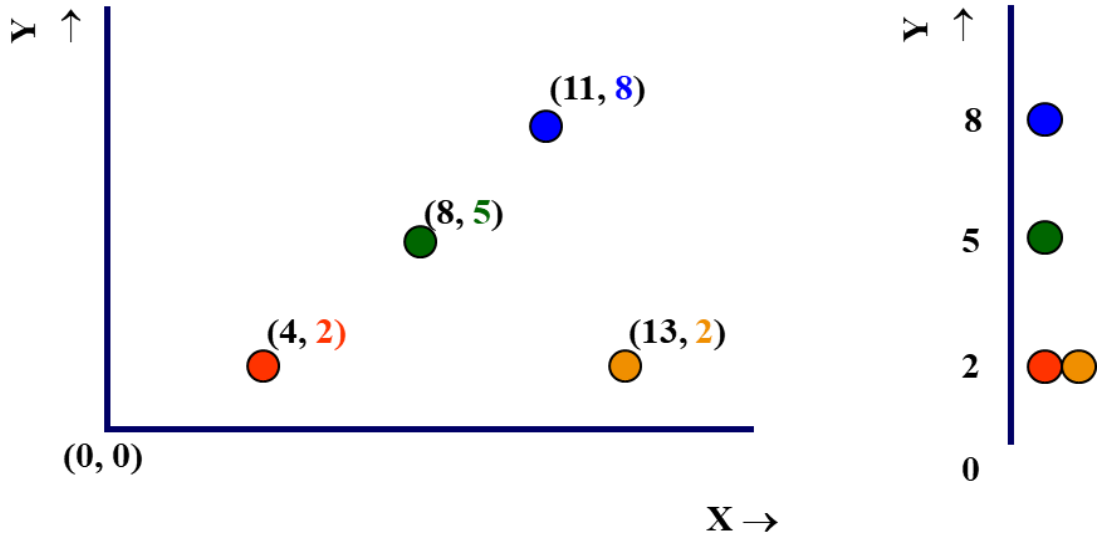
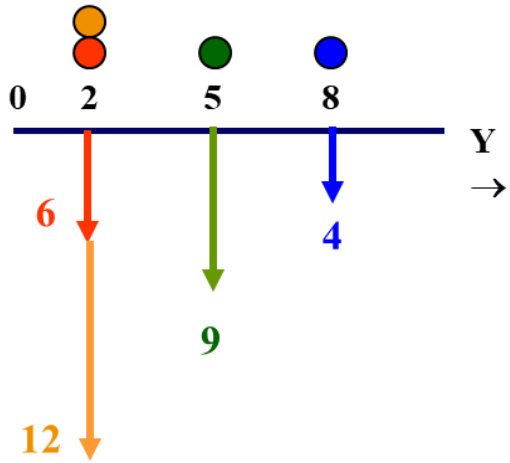
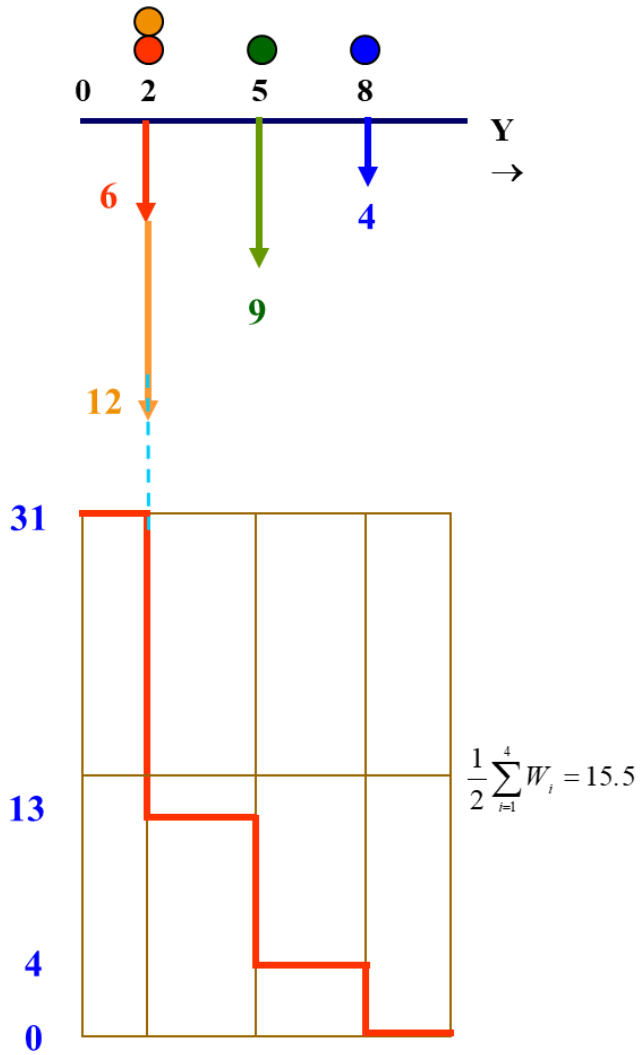
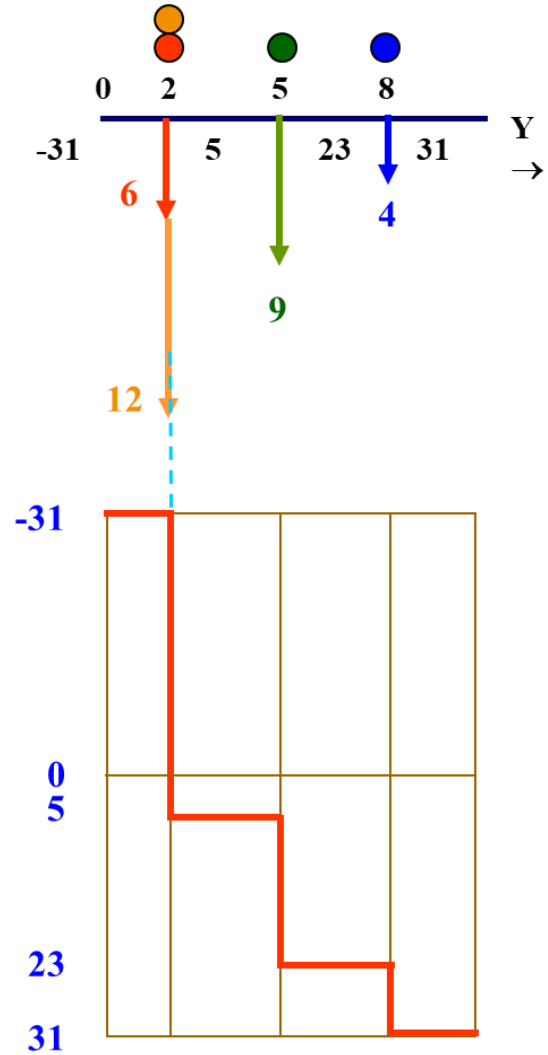




Cumulative Weight Diagram



Resultant Force Diagram



Reflection: Absolute vs Squared Distances

If we want to minimize **squared distances along one dimension**, then our goal is to minimize:

$$S(a) = \sum (x_i - a)^2$$

If we were to solve it:

$$\frac{dS}{da} = \sum 2(a - x_i) = 2na - 2\sum x_i = 0$$

$$a = \frac{\sum x_i}{n}$$

Reflection: Absolute vs Squared Distances

So, if we want to minimize **squared distances along one dimension**, then we are essentially solving for the mean.

Given information about the weights, this just becomes a weighted mean.

Simply, we can treat the weights (trips to a facility) as a new facility at that location.

Reflection: Absolute vs Squared Distances

If we want to minimize **absolute distances along one dimension**, then our goal is to minimize:

$$S(a) = \sum |x_i - a|$$

If we were to solve it, consider that:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\frac{\mathrm{d}}{\mathrm{dx}}|x| = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

$$\frac{dS}{da} = \#(x_i < a) - \#(x_i > a)$$

Reflection: Absolute vs Squared Distances

So, if we want to minimize **absolute distances along one dimension**, then we are essentially solving for the median.

Given information about the weights, then we can again treat the weights (trips to a facility) as a new facility at that location.

In-class Activity

There are five existing machines located at P1 (1,1), P2 (6,2), P3(2,8), P4 (3,6), P5 (8,4). The number of trips per day between the new machine tool and the existing machines are: 10, 20, 25, 20 and 25, respectively. Find the location of the new machine tool.

In-class Activity

Solution

Variables		Constraints		Objective
x	3	2>=	0	230
r1	2	0>=	0	
s1	0	0>=	0	
r2	0	3>=	0	
s2	3	1>=	0	
r3	1	0>=	0	
s3	0	0>=	0	
r4	0	0>=	0	
s4	0	0>=	0	
r5	0	5>=	0	
s5	5			
		1=	1	
		6=	6	
		2=	2	
		3=	3	
		8=	8	

In-class Activity

Solution

Variables		Constraints		Objective
y	4	3>=	0	210
r1	3	0>=	0	
s1	0	2>=	0	
r2	2	0>=	0	
s2	0	0>=	0	
r3	0	4>=	0	
s3	4	0>=	0	
r4	0	2>=	0	
s4	2	0>=	0	
r5	0	0>=	0	
s5	0			
		1=	1	
		2=	2	
		8=	8	
		6=	6	
		4=	4	